

Physical Computing with Arduino

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What is Physical Computing?

- **Physical Computing** is the design and creation of interactive systems that use hardware and software to sense and respond to the physical world. It bridges the gap between the digital world and the physical world.
 - E.g., A smart lamp that changes brightness based on ambient light.
- A physical computing system typically:
 - **Senses** the environment using sensors
 - **Processes** information using a microcontroller or computer
 - **Acts** on the environment using actuators
 - Input → Processing → Output

What is IoT?

- IoT = Physical Device + Connectivity + Data Exchange
- **Internet of Things (IoT)** refers to a network of physical devices ("things") that are equipped with sensors, actuators, processors, software, and communication capabilities, allowing them to collect data, exchange information, and interact with other devices or cloud services over a network (typically the Internet).

Feature	Physical Computing	IoT
Focus	Interaction with physical world	Connected smart systems
Internet Required	No	Usually Yes
Networking	Optional	Essential
Data Sharing	Local	Remote
Cloud Integration	Not required	Common
Mobile Access	Usually absent	Common
Scale	Single device	Multiple connected devices
Analytics	Local processing	Cloud analytics possible

What is Arduino?

<https://www.arduino.cc/>

- Arduino is an open-source electronics platform consisting of:
 - Hardware Boards
 - Software Development Environment (Arduino IDE)
- It was developed to make embedded systems and physical computing accessible to students, hobbyists, artists, and engineers.
- Arduino boards contain a microcontroller that can be programmed to read sensors and control actuators.

Arduino Hardware

Arduino Hardware

<https://www.arduino.cc/en/hardware/>

Family	Typical Use
Uno Series	Education, prototyping, general embedded systems
Nano Series	Compact embedded projects
Mega Series	Projects requiring many I/O pins
MKR Series	IoT and low-power applications
GIGA Series	High-performance embedded systems
Portenta Series	Industrial IoT, Edge AI
Nicla Series	Tiny sensor and AI nodes
UNO Q Series	Linux + AI + Edge Computing
Legacy Boards	Uno R3, Leonardo, Due, Zero, etc.

Arduino UNO Family

- **Arduino Uno R3:**

- The board that made Arduino famous.
- Learning embedded systems, basic sensor projects, electronics education

- **Arduino UNO R4 Minima:**

- Released in 2023 as the successor to the Uno R3.

- **Arduino UNO R4 WiFi:**

- R4 Minima+
- IoT, Smart home projects, Cloud-connected systems

			
Arduino UNO Q	Arduino UNO R4 WiFi	Arduino UNO R4 Minima	Arduino UNO R3
			
Arduino UNO Mini Limited Edition	Arduino UNO WiFi Rev2	Arduino UNO R3 SMD	

- **Arduino UNO Q:**

- Introduced in 2025/2026 as a major step beyond traditional microcontroller boards.
- For: AI at the edge, Vision systems, Robotics, Industrial automation, Linux-based embedded applications
- Between an Arduino and a Raspberry Pi.

Arduino UNO Shields

- A **shield** is an add-on board that plugs directly onto an Arduino board (typically Uno, Mega, Due, or R4) to add new functionality.
- A hardware module that extends the capabilities of a microcontroller board.



Arduino Motor
Shield Rev3



Arduino 4 Relays
Shield



Arduino Ethernet
Shield Rev2



Arduino 9 Axis
Motion Shield



Arduino UNO SPE
Shield

Arduino Nano Family

<https://store.arduino.cc/pages/nano-family>

- Small form-factor boards ideal for embedded products.
- **Arduino Nano:**
 - Classic, breadboard-friendly
- **Arduino Nano R4:**
 - Latest Nano
- **Arduino Nano 33 IoT**
 - IoT ready

			
Arduino Nano 33 IoT	Arduino Nano RP2040 Connect	Arduino Nano ESP32	Arduino Nano 33 BLE Sense
			
Arduino Nano 33 BLE	Arduino Nano Every	Arduino Nano	Arduino Nano Motor Carrier
			
Nano Connector Carrier			

	Arduino Nano	Nano ESP32-S3	Nano 33 IoT	Nano 33 BLE	Nano 33 BLE Sense Rev2	Nano RP2040 Connect	Nano Every	Nano Matter	Nano R4
Microcontroller	ATmega328	ESP32-S3	SAMD21 Arm® Cortex®-M0+ / u-blox® NINA-W102	nRF52840 / u-blox® NINA-B306	nRF5284 /u-blox® NINA-B306	Raspberry Pi RP2040 /u-blox® NINA-W102	ATMega4809	Silicon Labs MGM24	R7FA4M1AB3CFM
USB connector	Mini-B USB	USB-C®	Micro USB	Micro USB	Micro USB	Micro USB	Micro USB	USB-C®	USB-C®
I/O voltage	5 V	3.3 V	3.3 V	3.3 V	3.3 V	3.3 V	5 V	3.3 V	5 V
Input range	7-12 V	5-21 V	7-12 V	7-12 V	7-12 V	5-21 V	7-12 V	5-21 V	6-21 V
Clock speed	16 MHz	up to 240 MHz	48 MHz	64 MHz	64 MHz	133 MHz	16 MHz	78 MHz	48 MHz
SRAM	2 kB	512 kB	256 kB	256 kB	256 kB	264 kB	6 kB	256 kB	32 kB
Flash	32 kB	16 MB	1 MB	1 MB	1 MB	16 MB	48 kB	1536 kB	256 kB
Wi-Fi®	-	YES	YES	-	-	YES	-	-	-
Bluetooth® LE	-	YES	YES	YES	YES	YES	-	YES	-
Digital inputs	20	22	22	14	14	22	22	22	22
Analog inputs	8	8	8	8	8	8	8	8	8
ADC resolution	10 bits	12 bits	12 bits	12 bits	12 bits	12 bits	10 bits	12 bits	14-bits max
PWM	6	20	11	14	14	18	5	20	6

Arduino Mega Family

- In the Mega family, you will find the boards for the projects that demand lots of computing power and GPIO pins.
- Arduino Mega 2560 Rev3:
 - ATmega2560, 54 digital pins, 16 analog inputs
 - CNC controllers, 3D printers, Robotics, Industrial control



Arduino Mega 2560
Rev3



Arduino Due



Arduino GIGA R1
WiFi



Arduino GIGA
Display Shield

Arduino MKR Family

<https://store.arduino.cc/collections/mkr-family>

- Designed specifically for IoT.
- Enables Wi-Fi®, Bluetooth®, LoRa®, Sigfox®, NB-IoT communication.
- Low power processor, and are equipped with a crypto chip for secure communication.



Arduino MKR 1000 WiFi



Arduino MKR WiFi 1010



Arduino MKR FOX 1200



Arduino MKR WAN 1300



Arduino MKR WAN 1310



Arduino MKR GSM 1400



Arduino MKR NB 1500



Arduino MKR Vidor 4000



Arduino MKR Zero

Arduino Modulino

- Small modules that connect to your board via Qwiic and bring plug-and-play sensors, actuators & more to your projects.

			
Modulino Buttons	Modulino Buzzer	Modulino Distance	Modulino Knob
			
Modulino Movement	Modulino Pixels	Modulino Thermo	Modulino Joystick
			
Modulino Light	Modulino Latch Relay	Modulino Vibro	Modulino LED Matrix

Retired Arduino Products

<https://docs.arduino.cc/retired/>

Boards

Arduino 101
Arduino BT (Bluetooth)
Arduino BT v1
Arduino Diecimila
Arduino Duemilanove
Arduino Esplora
Arduino Ethernet Rev3 with PoE
Arduino Ethernet Rev3 without PoE
Arduino Fio
Arduino Gemma
Arduino Industrial 101
Arduino ISP
Arduino Leonardo ETH
Arduino Leonardo ETH 2 with PoE
Arduino M0

Shields

Arduino Ethernet Shield 2 with PoE
Arduino Ethernet Shield
Arduino GSM Shield
Arduino GSM Shield 2 (Antenna Connector)
Arduino GSM Shield 2 (Integrated Antenna)
Arduino Lucky Shield
Arduino USB Host Shield
Arduino WiFi Shield
Arduino WiFi Shield 101
Arduino Wireless Proto Shield
Arduino Wireless SD Shield
Arduino Xbee Shield
Arduino Yún Shield

Kits

Arduino IoT Prime
Arduino Basic Kit
Arduino Engineering Kit
Arduino On Android Kit
Arduino Proto Extension Kit
Arduino Pro Gateway
Shield - MEGA Proto KIT Rev3

Arduino Software

Arduino Software

<https://www.arduino.cc/en/software/>

- Arduino is no longer just the Arduino IDE. The modern ecosystem includes graphical programming tools, cloud-based development, MicroPython support, Linux application development for UNO Q, command-line tools, and professional deployment platforms.

Arduino IDE (v2.3.10 by June 2026)

- The primary development environment for Arduino boards.
- Features: C/C++ programming, Code editor with IntelliSense, Library Manager, Board Manager, Serial Monitor, Serial Plotter, Debugging support, Integrated compiler and uploader
- The standard platform for teaching embedded systems and physical computing.

Arduino Cloud

<https://cloud.arduino.cc/>

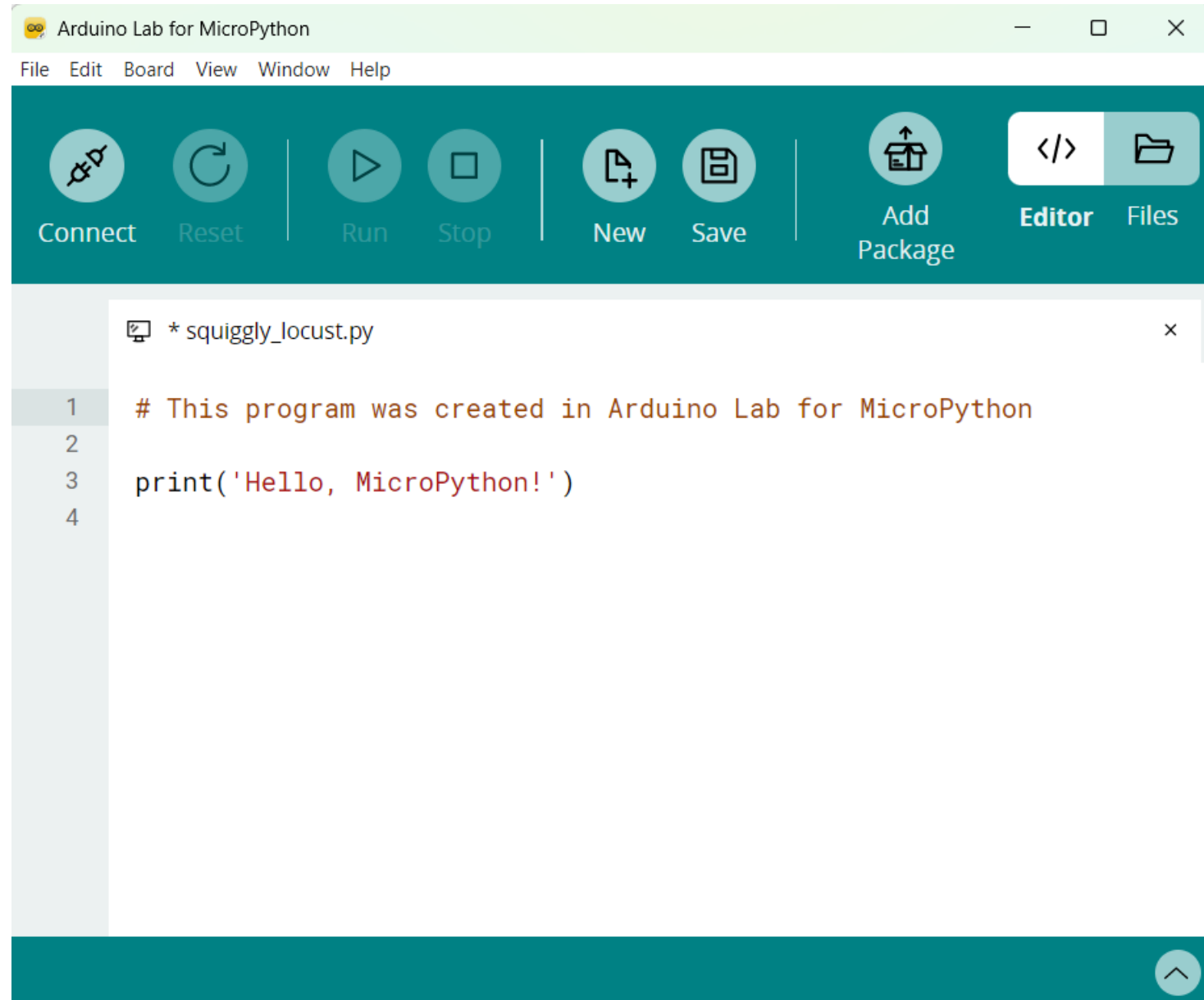
- Arduino's cloud platform for IoT development.
- Features: Browser-based coding, Device management, OTA (Over-The-Air) updates, Dashboards, Data logging, Remote monitoring, Mobile app integration
- Widely used in IoT courses and smart agriculture projects.
- Works only with compatible boards (Uno R3 not supported).

Arduino App Lab

- App Lab is one of Arduino's newest software platforms, developed especially for the **Linux-based UNO Q** ecosystem.
- Instead of programming only microcontrollers, App Lab allows development of Linux applications, AI applications, Computer vision applications, Edge computing software, Graphical interfaces.

Arduino Lab for MicroPython

- Features: Python programming, Interactive REPL, File management, Firmware installation, Serial terminal.
- Works only with compatible boards (Uno R3 not supported).
- Another option: [CircuitPython](#)



Arduino CLI

- Command-line development environment.
- Useful for Automation, CI/CD pipelines, Large projects, Professional workflows.
- Often used together with VS Code.

VS Code + Arduino Extension

- Using **VS Code** with the Arduino extension provides a more professional development environment than the standard Arduino IDE, especially for larger projects involving Arduino, ESP32, STM32, and IoT applications.
- Difficult to configure.

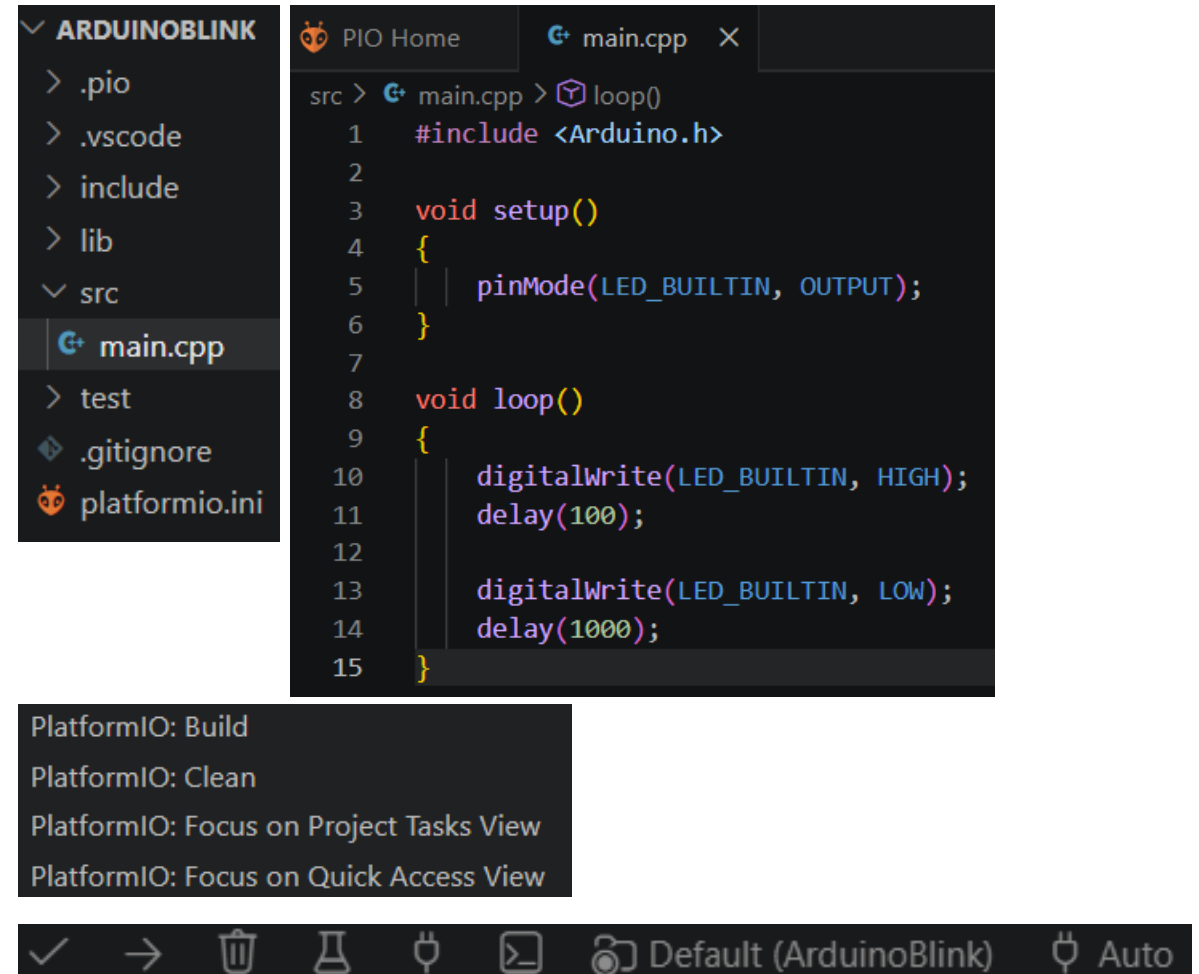
Arduino IDE	VS Code
Simple	Advanced
Single-file sketches	Multi-file projects
Basic editor	Powerful editor
Limited code navigation	IntelliSense
Basic debugging	Advanced debugging
Small projects	Large projects
Beginner-friendly	Professional workflow

VS Code + PlatformIO Extension

- **PlatformIO is often the easiest and most reliable way to program an Arduino Uno R3 from VS Code.** Unlike the Arduino extension, PlatformIO automatically downloads the AVR compiler, libraries, upload tools, and board definitions.

Steps

- Install VS Code (v1.125.1 as June 2026).
- Search for **PlatformIO IDE** extension and Install it.
- Create an Arduino Uno Project
 - Add code to src/main.cpp
- Build and Upload
 - Commands: PlatformIO: Build and PlatformIO: Upload
 - Or using the icons.



Arduino Using Blockly Visual Programming

<https://www.blockly.com/>

- An open-source, flexible library for developers to build visual programming editors with drag-and-drop blocks – powering the world’s most popular block-based coding platforms.
 - E.g., LED Blink demo:
<https://blocklyduino.github.io/BlocklyDuino/blockly/apps/blocklyduino/>
- Another similar platform:
[**Open Roberta Lab**](#)

[BlocklyDuino](#) > web-based visual programming editor for arduino



Arduino Using ThinkerCad Circuits

<https://www.tinkercad.com/circuits>

- **Tinkercad Circuits** is a free web-based electronics and Arduino **simulation environment** developed by Autodesk. It allows students, hobbyists, and educators to design, build, and simulate electronic circuits without requiring physical hardware.
- Another option: [Wokwi](#)



Code

Start Simulation

Send To



```
// C++ code
//
void setup()
{
  pinMode(LED_BUILTIN, OUTPUT);
}

void loop()
{
  digitalWrite(LED_BUILTIN, HIGH);
  delay(1000); // Wait for 1000 millisecond(s)
  digitalWrite(LED_BUILTIN, LOW);
  delay(1000); // Wait for 1000 millisecond(s)
}
```

Components
Basic

Search



Resistor



LED



Pushbutton



Potentiometer



Capacitor



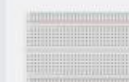
Slideswitch



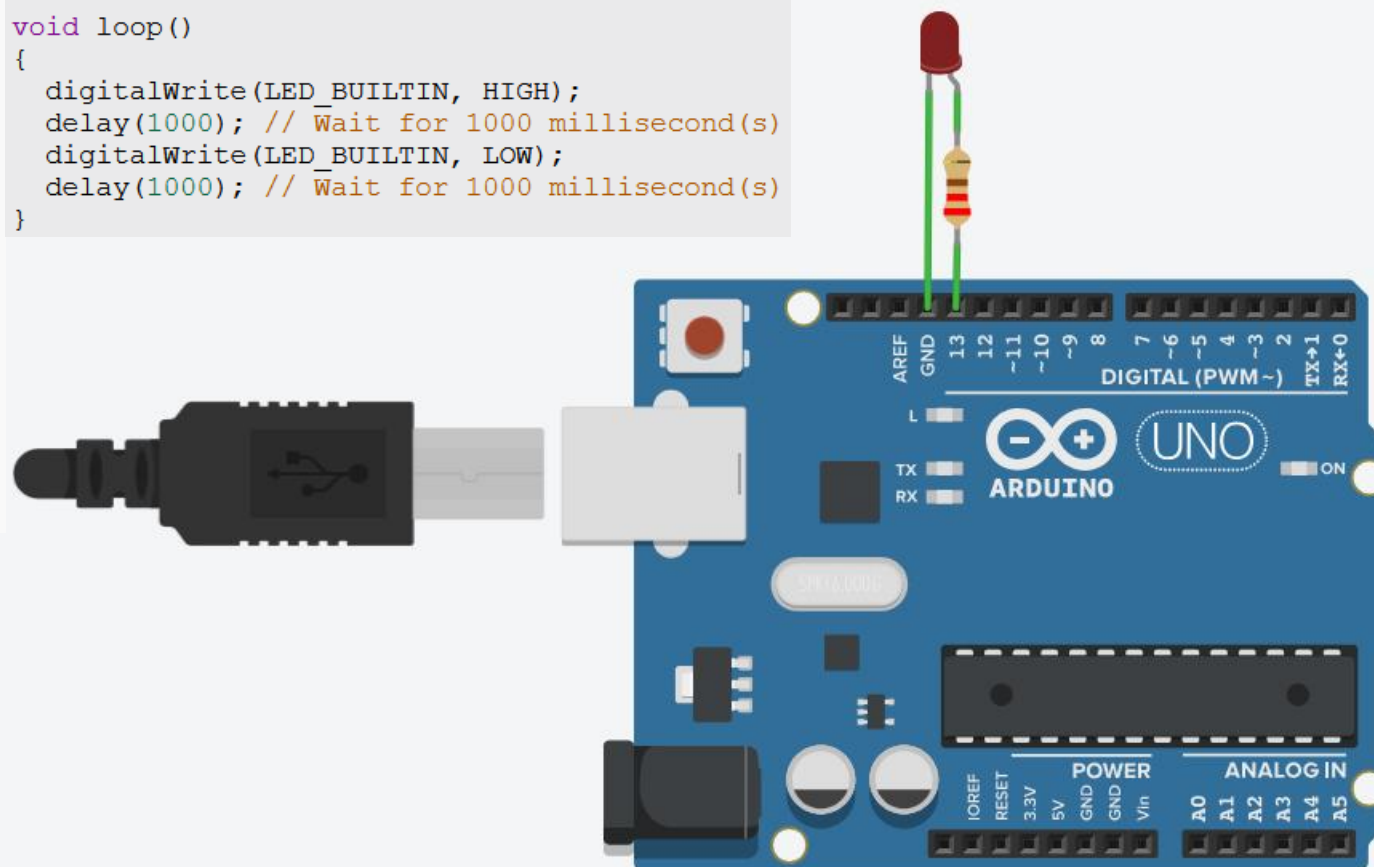
9V Battery

Coin Cell 3V
Battery

1.5V Battery

Breadboard
Small

micro:bit

Arduino Uno
R3

Arduino Community

Arduino Community

- The **Arduino Community** is one of the largest and most influential open-source hardware and physical computing communities in the world. It consists of students, hobbyists, educators, researchers, makers, engineers, artists, and companies who use and contribute to the Arduino ecosystem.
- The success of Arduino is due not only to its hardware and software, but also to its global community that continuously develops libraries, tutorials, projects, and educational resources.

Components of the Arduino Community

- Official Arduino Community
 - Maintained through the [Arduino Official Website](#)
 - Provides: Documentation, Hardware information, Software downloads, Educational resources, Project tutorials
 - [Arduino Forum](#)
- Open-Source Contributors
 - The source code is publicly available through [Arduino GitHub Organization](#)
 - Thousands of developers contribute: Libraries, Hardware designs, Examples, Bug fixes, Documentation
- Educational Community
 - Many universities use Arduino as the first platform for teaching embedded systems and physical computing.
- Makers and Hobbyists, Research Community, User Groups and Meetups

<https://projecthub.arduino.cc/>



Audio and sound



Internet of things



Installations



Home
automation



Flying things



Lab tools



Environment
monitoring



Robotics



Interactive
games



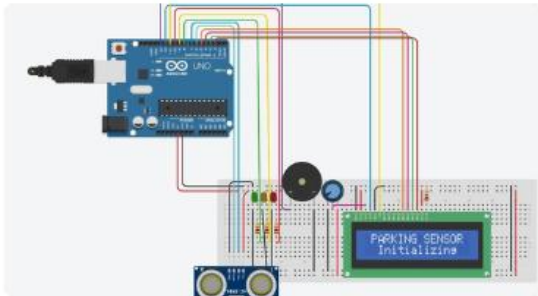
What do you want to make?

All products

Any type

Any difficulty

6,199 projects · Trending



Parking Sensor Distance Detector

Tutorial by [wonderfulviking](#)

4143 · 0 · 5



Voice-Controlled LED Matrix Display using Arduino Uno Q

Tutorial by [vijaybarkade](#)

770 · 0 · 1



Build a smart mirror with an Arduino UNO Q

Showcase by [kamitronix](#)

8458 · 1 · 10



Slot Machine Treat Dispenser

Teardown by [circuitcindy](#)

5446 · 1 · 6

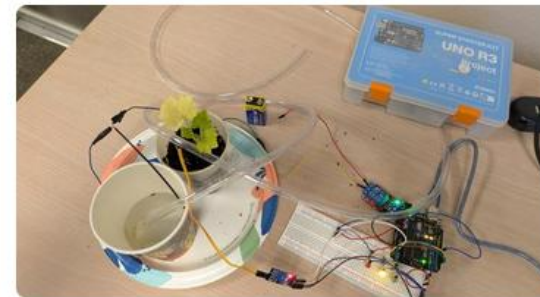


UNO Q Braccio: Edge Impulse + ROS 2 Controlled Robot Arm wit...



GigaNES

Hiran Ekanayake @ UCSC



Automated Irrigation System / Soil Moisture Monitoring System



Ripple - My desktop widget

Arduino Programming Activities

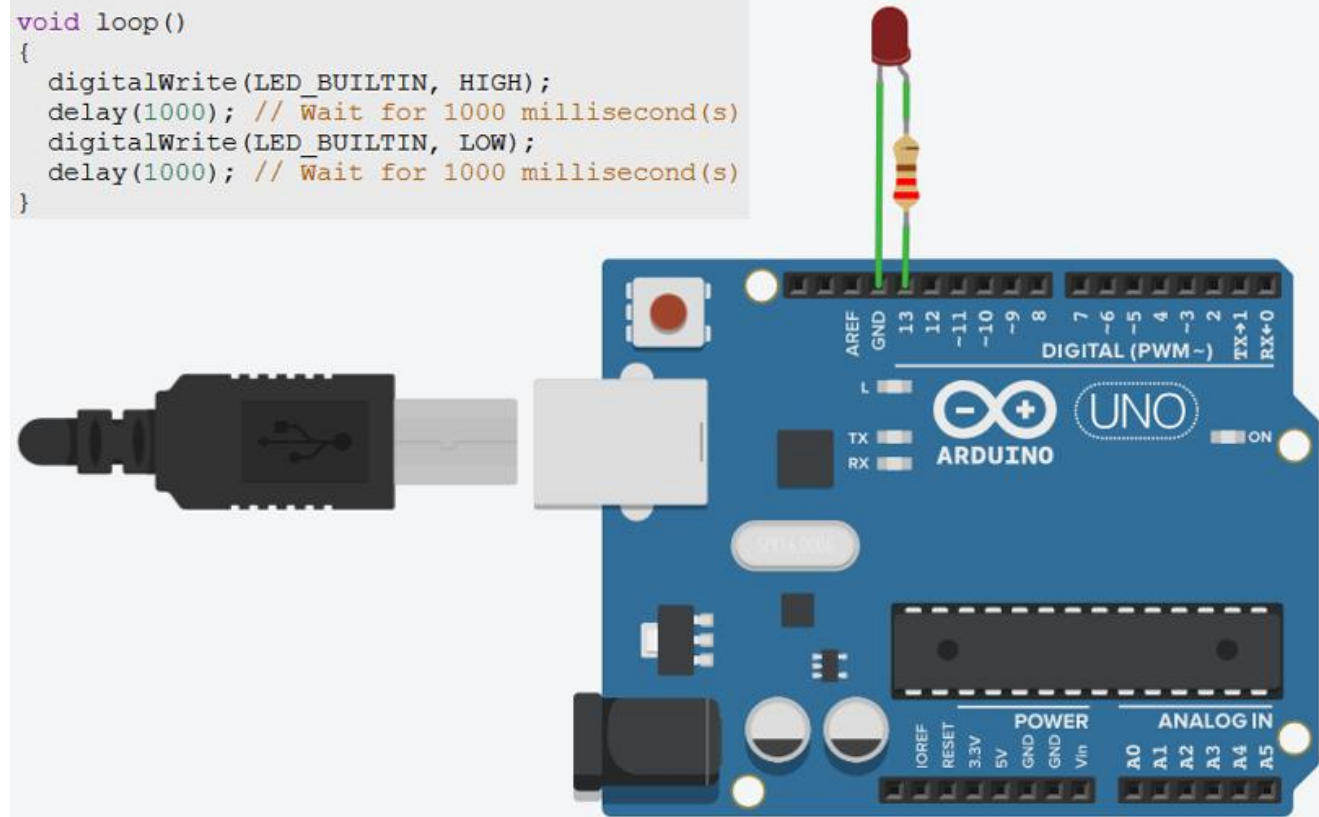
Conduct self-directed exploration of the assigned laboratory exercises using Arduino hardware or a suitable simulator. Submit a report containing the circuit designs, experimental setups, relevant screenshots or photographs, Arduino sketches, execution results, and a brief discussion of observations and conclusions.

Module 1: Introduction to Arduino and Embedded Systems

- **Lab 1: LED Blink**
- **Concepts**
 - Embedded program structure
 - setup() and loop()
 - Digital output
 - Timing using delay()
- **Tasks**
 - Blink one LED.
 - Blink multiple LEDs.
 - Create different blink patterns.

```
// C++ code
//
void setup()
{
  pinMode(LED_BUILTIN, OUTPUT);
}

void loop()
{
  digitalWrite(LED_BUILTIN, HIGH);
  delay(1000); // Wait for 1000 millisecond(s)
  digitalWrite(LED_BUILTIN, LOW);
  delay(1000); // Wait for 1000 millisecond(s)
}
```



Module 1: Introduction to Arduino and Embedded Systems Cont.

- **Lab 2: Traffic Light Controller**

- **Concepts**

- State-based control
- Sequential logic
- Digital outputs

- **Tasks**

- Implement: Red → Green → Yellow → Red
- Extension: Pedestrian crossing signal.

Module 2: Digital Inputs

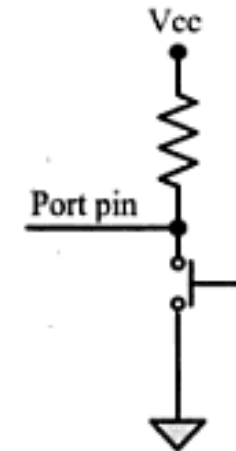
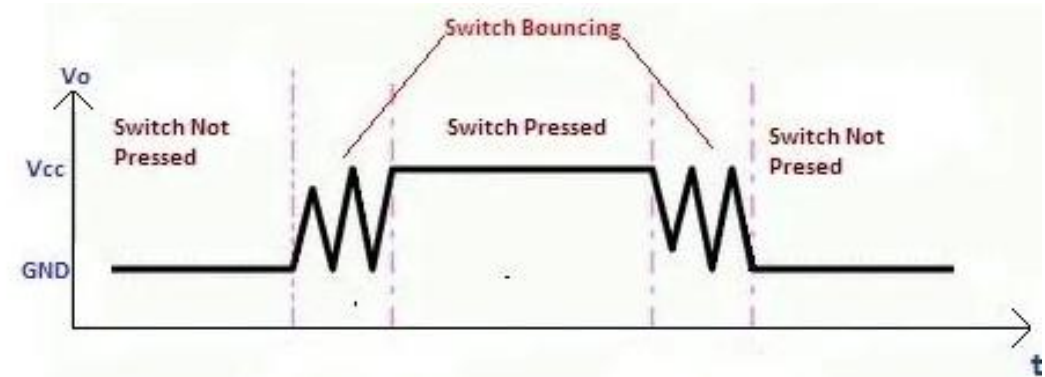
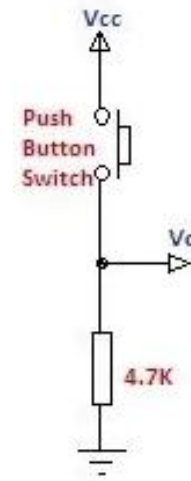
- **Lab 3: Push Button Interface**

- **Concepts**

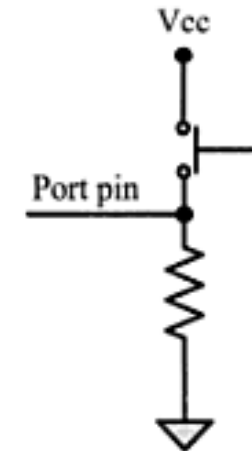
- Digital input
- Pull-up and pull-down resistors
- Debouncing

- **Tasks**

- LED ON while button pressed.
- Toggle LED on button press.
- Count button presses.



'LOW' Pulse generator



'HIGH' Pulse generator

Module 2: Digital Inputs

- **Lab 4: Electronic Voting Counter**
- **Concepts**
 - Event detection
 - Counters
 - State management
- **Tasks**
 - Display vote counts through Serial Monitor.

Module 3: Analog Inputs

- **Lab 5: Potentiometer-Based LED Dimmer**

- **Concepts**

- ADC
- Analog signals
- PWM

- **Tasks**

- Control LED brightness using a potentiometer.

Module 3: Analog Inputs

- **Lab 6: Light Intensity Monitoring**
- **Components**
 - LDR
- **Concepts**
 - Sensor calibration
 - Analog measurement
- **Tasks**
 - Display light levels in Serial Monitor.

Module 4: Serial Communication

- **Lab 7: Serial Monitor Interaction**
- **Concepts**
 - UART
 - Serial communication
- **Tasks**
 - Create a menu-driven application: 1 -> LED ON, 2 -> LED OFF, 3 -> Blink

Module 4: Serial Communication Cont.

- **Lab 8: Serial Sensor Dashboard**
- **Concepts**
 - Data logging
 - Human-computer interaction
- **Tasks**
 - Transmit sensor readings continuously.

Module 5: Sensors

- **Lab 9: Temperature and Humidity Monitoring**

- **Components**

- DHT22

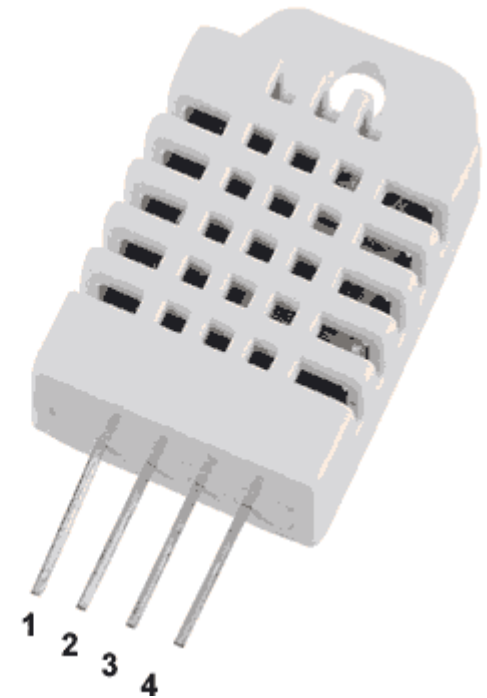
- **Concepts**

- Digital sensors
- Library usage

- **Tasks**

- Display: Temperature & Humidity

DHT22 pins	
1	VCC
2	DATA
3	NC
4	GND



Module 5: Sensors Cont.

- **Lab 10: Distance Measurement System**
- **Components**
 - HC-SR04
- **Concepts**
 - Time-of-flight sensing
- **Tasks**
 - Measure distance and generate warnings.



Module 5: Sensors Cont.

- **Lab 11: Motion Detection Alarm**
- **Components**
 - PIR sensor
- **Concepts**
 - Event-driven systems
- **Tasks**
 - Trigger alarm when motion detected.



Module 6: Actuators

- **Lab 12: Buzzer Alert System**
- **Concepts**
 - Tone generation
- **Tasks**
 - Generate alarm sounds
 - Generate musical notes



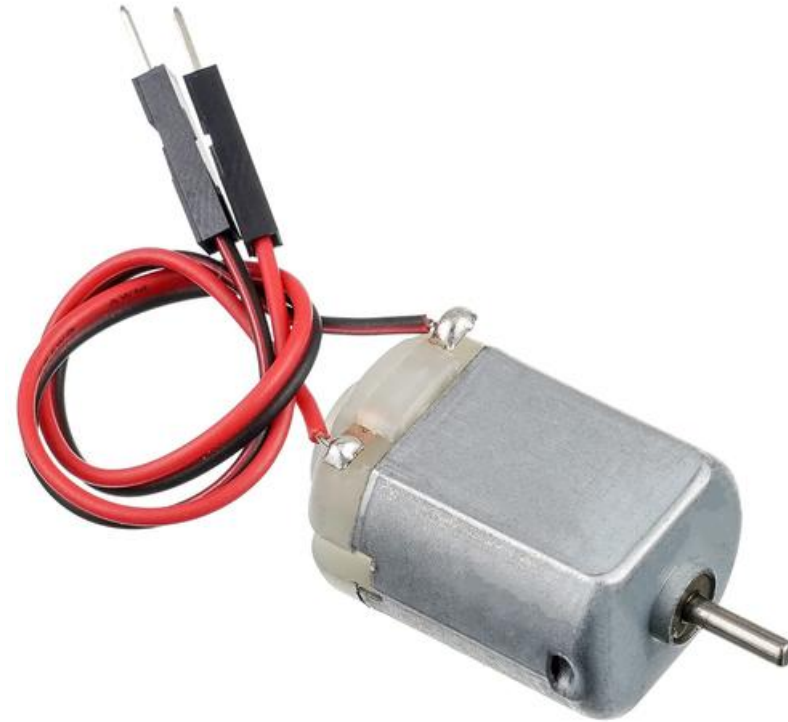
Module 6: Actuators Cont.

- **Lab 13: Servo Motor Control**
- **Concepts**
 - PWM
 - Position control
- **Tasks**
 - Control servo angle using potentiometer
 - Control servo angle using serial commands



Module 6: Actuators Cont.

- **Lab 14: DC Motor Speed Controller**
- **Concepts**
 - PWM
 - Motor drivers
- **Tasks**
 - Control speed and direction.



Module 7: LCD and User Interfaces

- **Lab 15: LCD Sensor Display**
- **Components**
 - 16×2 LCD
- **Concepts**
 - User interfaces
- **Tasks**
 - Display sensor readings.



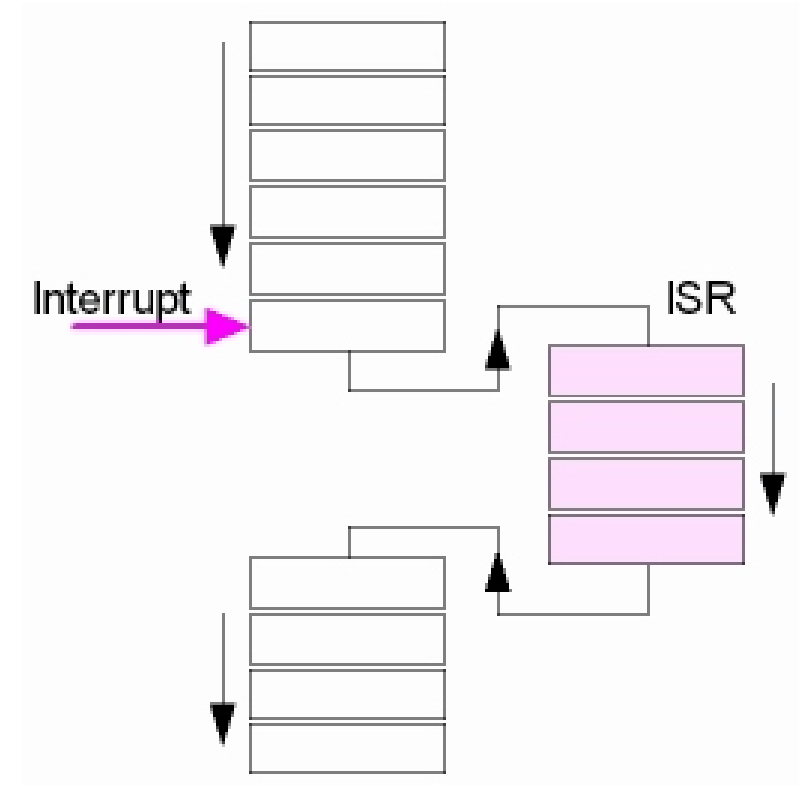
Module 7: LCD and User Interfaces Cont.

- **Lab 16: Embedded Menu System**
- **Concepts**
 - User interaction
 - State machines
- **Tasks**
 - Navigate menu using buttons.



Module 8: Interrupts

- **Lab 17: Interrupt-Based Event Counter**
- **Concepts**
 - Hardware interrupts
 - ISR design
- **Tasks**
 - Count external events. E.g., wheel rotation, object detection



Module 8: Interrupts Cont.

- **Lab 18: Reaction Time Measurement**
- **Concepts**
 - Precise timing
 - Interrupts
- **Tasks**
 - Measure user reaction time.

Module 9: Timers and Real-Time Systems

- **Lab 19: Non-Blocking Traffic Light**
- **Concepts**
 - millis()
 - Cooperative multitasking
- **Tasks**
 - Replace delay() with timer-based design.

Module 9: Timers and Real-Time Systems Cont.

- **Lab 20: Multi-Task Embedded System**
- **Concepts**
 - Concurrent tasks
- **Tasks**
 - Simultaneously read sensor, blink LED, and update display

Module 10: Data Acquisition and Logging

- **Lab 21: Environmental Data Logger**

- **Components**

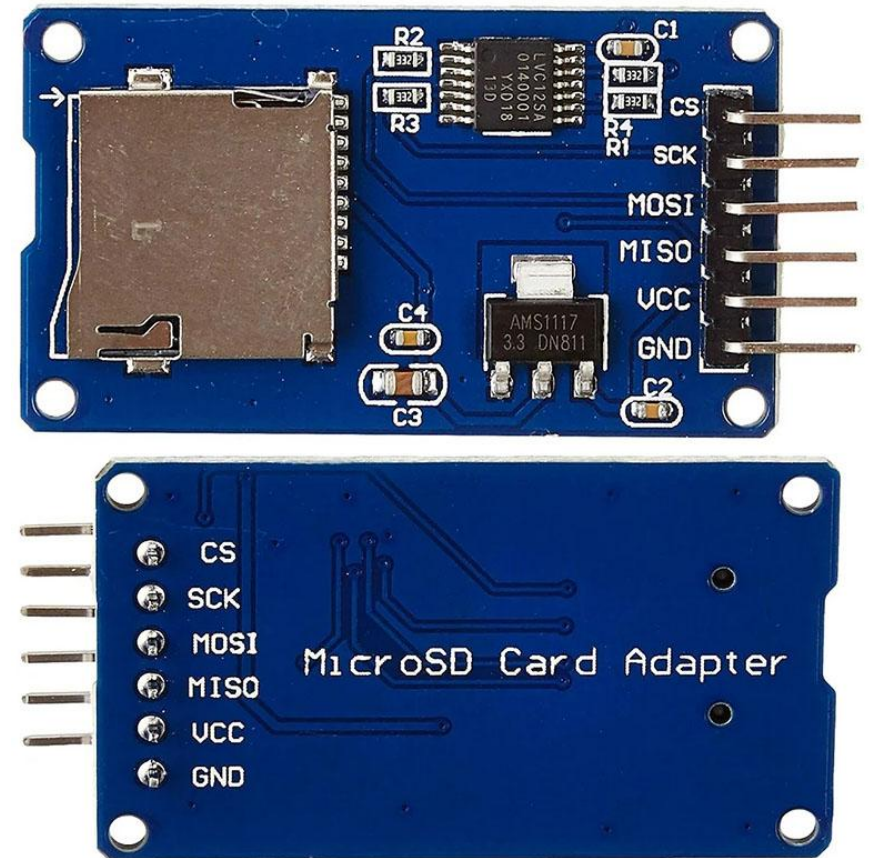
- DHT22
- SD Card

- **Concepts**

- File systems
- Data logging

- **Tasks**

- Store readings with timestamps.



Module 10: Data Acquisition and Logging Cont.

- **Lab 22: Sensor Data Analysis**

- **Concepts**

- Data collection
- Visualization

- **Tasks**

- Export data to CSV.
- Analyze in Excel or Python

Module 11: Communication Protocols

- **Lab 23: I2C Communication**

- **Components**

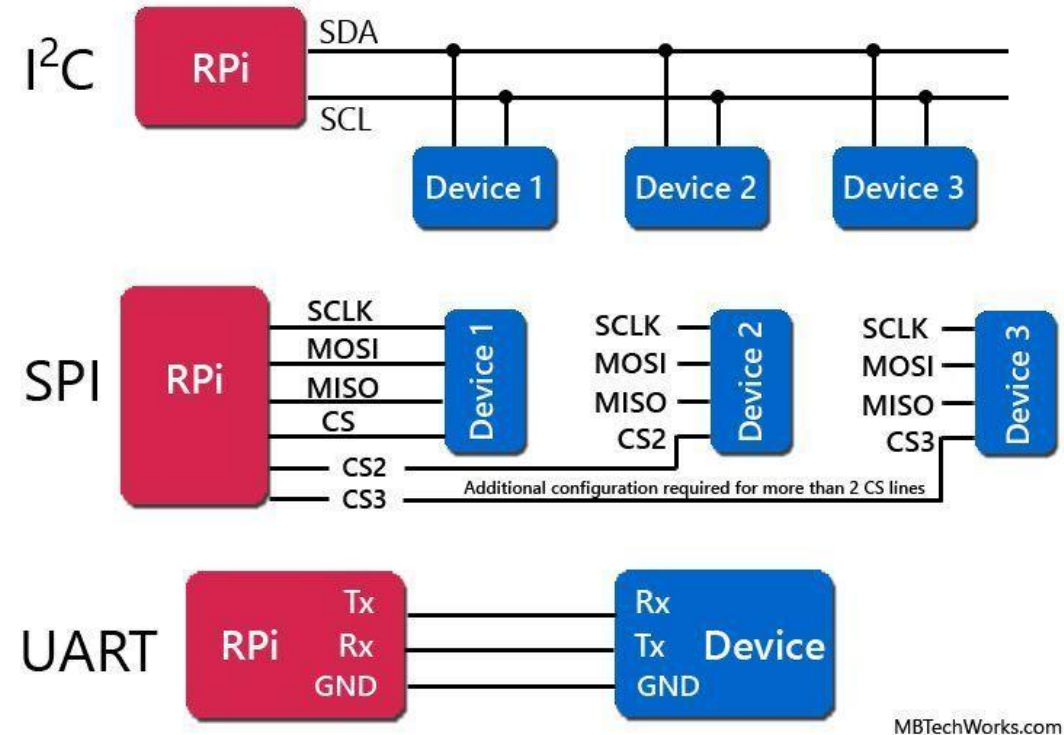
- LCD
- RTC

- **Concepts**

- I2C protocol

- **Tasks**

- Read and display time.



Module 11: Communication Protocols Cont.

- **Lab 24: SPI Communication**
- **Components**
 - SD Card
- **Concepts**
 - SPI protocol
- **Tasks**
 - Store and retrieve data.